# Neural Heterogeneity of Word Response

Proposal?

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## Gap

Studies of free recall describe the average response to a word, with high gamma rising first in visual regions and later in prefrontal regions (Burke et al., 2014; Kucewicz et al., 2019), and theta, alpha/beta and gamma power largely rising at separate sites, in separate sequences (Marks et al., 2021). These studies average over word events and rarely account for the heterogeneity across electrodes in a region, or other factors contributing towards variability in responses. How much of the response differs between subjects, electrodes, sessions and word events, which part is a stable property of a site, and what changes it from one word event to the next have not been measured. With 361 subjects, 1007 sessions and ∼200 words per session, this dataset can answer these questions. The contribution of task structure towards the memory-related component has been identified by DeHaan and Halpern (TBD), which the main mixed-effects model used here draws from.

## Aims

**Aim 1. How much of the response is stable, and how much changes from one word to the next?** Split the response to each word event (word on vs. word off, Cohen's $d$ effect size) into subject, electrode, session, task and word-event components per region, with a random-intercept model. *Preliminary (up to 5 electrodes per subject and ROI; Figs. 1B, 3B):* most variance across word events is unique to an electrode during a word event (alpha 71–86%, high gamma 56–89%, latency 82–94%). The next largest part is a word event's deviation shared by all of an ROI's electrodes (alpha 11–25%, high gamma 6–15%, latency 4–13%). Session, list in session and task structure are each small (≤3%). Stable differences are largest in occipital high gamma (subject 8%, electrode 22%, electrode in session 3%); for alpha outside occipital regions subject and electrode together are ≤2%, and for latency ≤6%.

Planned: the same split for power in the pre-stimulus blank alone, as a control (Fig. 5C).

**Aim 2. Which electrodes respond in which band, and do they covary?** Outside occipital regions 68–76% of electrodes show neither an alpha nor a high-gamma response, against 35% in occipital regions (paired $t$ over word events, FDR 0.05); 31% of occipital electrodes respond in both bands, 5–11% elsewhere. Which electrodes respond strongly within a band replicates between session halves (Fig. 2A; median $r = 0.90$ in occipital high gamma, 0.80 in occipital alpha, 0.20–0.59 elsewhere). Across electrodes the bands share the size of the response more than its direction (Fig. 2B): $r$ of $|d|$ is 0.46 in occipital regions and 0.13–0.21 elsewhere, while $r$ of signed

$d$ is between −0.11 and 0.15. In occipital regions, where high gamma rises, alpha falls at 19% of electrodes and rises at 8%.

Planned: does an electrode's latency replicate, also the anatomy of these response types within a region.

**Aim 3. What are the word-to-word effects?** Task structure effects are small next to the variability of single word events (Aim 1), but are still clear when pooled across events. Along the list high gamma falls in every ROI (Fig. 4; −0.004 to −0.012 $d$ per position) and alpha rises in every ROI but the hippocampus (0.004–0.005 $d$ per position). Across lists alpha rises in frontal, temporal, occipital and limbic regions, and high gamma has no trend. Latency shortens along the list only in frontal and limbic regions (0.7–1 ms per position), with no trend across lists (per-subject slopes vs. 0, FDR 0.05). Marks et al. (2021) found the bands at separate sites; at the same electrode they are also nearly independent across word events (Fig. 2C; subjects' mean $r$ between −0.01 and 0.11).

Planned: what the shared word-event component is (arousal, attention, or a common reference), the blank before a word against the neighbouring words (recovery from adaptation, Groen et al., 2022, vs. temporal expectation; Fig. 5), and latency per word event.

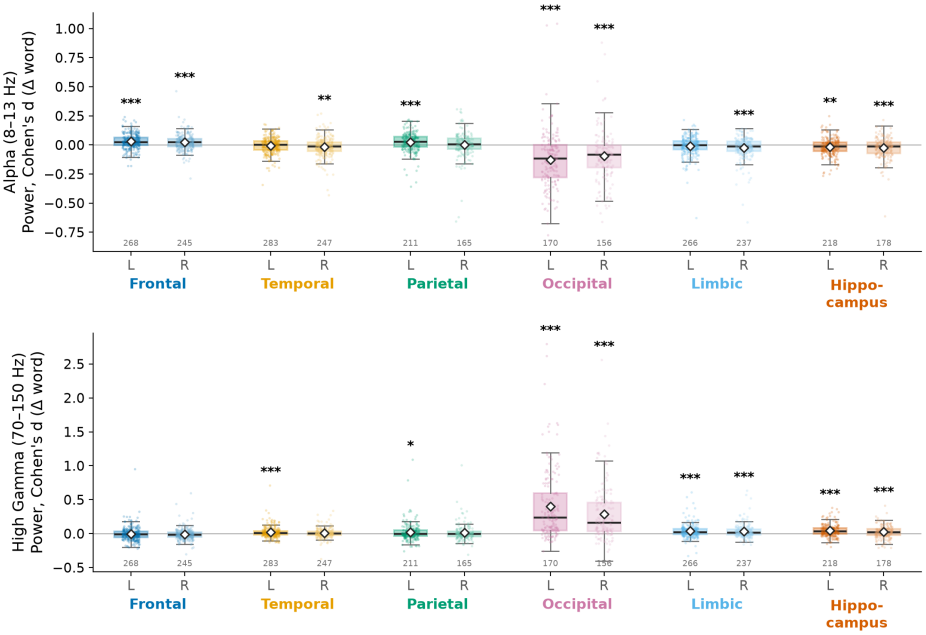
**Aim 4. Does connectivity between regions vary in the same way?** At word onset alpha ciPLV rises in 9 of 12 regions, with hubs first in occipital regions (0–400 ms) and then in limbic, parietal, frontal and temporal regions (Fig. 6A). Connectivity is less reliable across sessions than power (median $r$ 0.19 for alpha ciPLV and 0.14 for high-gamma AEC-c, vs. 0.67 and 0.81 for power; Fig. 6B).

Planned: the variance split of Aim 1 for synchrony (Fig. 6C).

Differences between locations are stable but large only in occipital high gamma; elsewhere most of the variance is from one word event to the next.

## Figures

**A** (Done)




**B** (Prelim, 5 electrodes per subject/ROI)

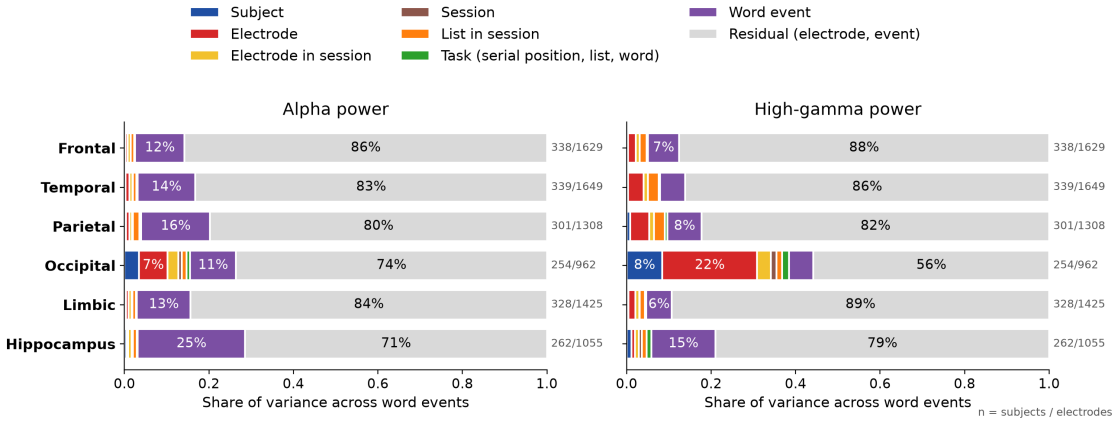
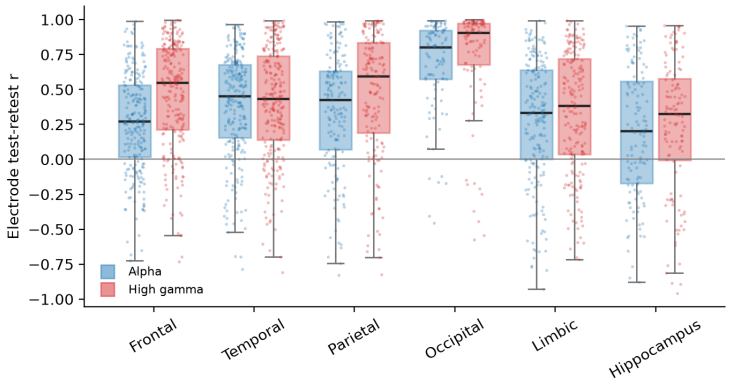
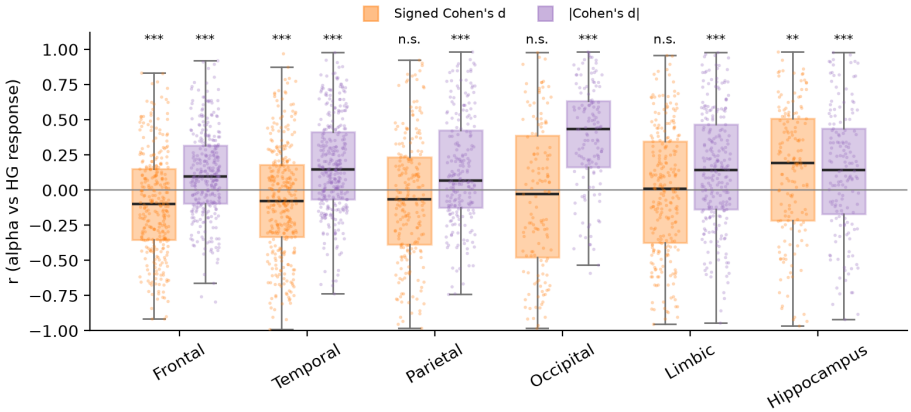



**Figure 1:** (A) Power change at word onset (Cohen's $d$, word on vs. off, across subjects). (B) Variance components of the response to each word event, one mixed model per ROI (Aim 1).

**A** (Done)




**B** (Done)




**C** (Done)

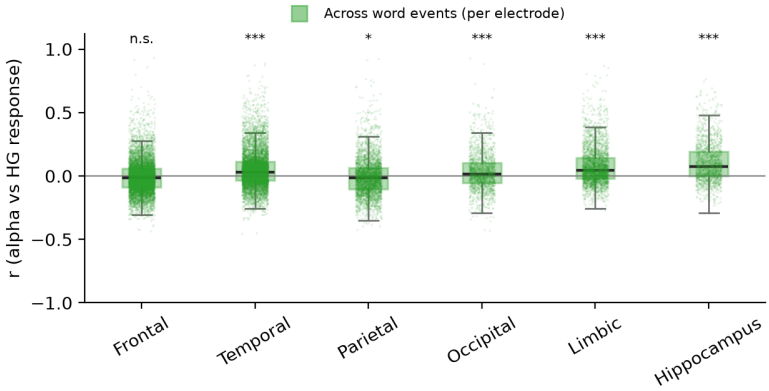
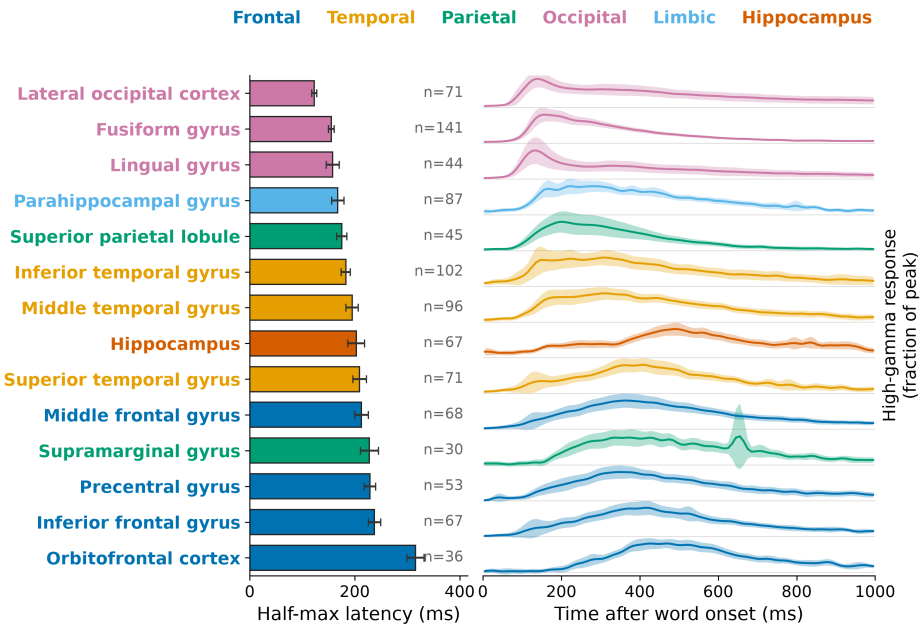



**Figure 2:** (A) Electrode test-retest: $r$ of electrodes' Cohen's $d$ between session halves (1 dot = 1 subject). (B) Alpha vs. high gamma across an ROI's electrodes: $r$ of Cohen's $d$ and of $|d|$ per subject (dots); stars: subjects' mean $r$ vs. 0, FDR. (C) Alpha vs. high gamma across word events: $r$ per electrode (dots); stars: subjects' mean $r$ vs. 0, FDR.

**A** (Done)




**B** (Prelim, 5 responsive electrodes per subject/ROI)

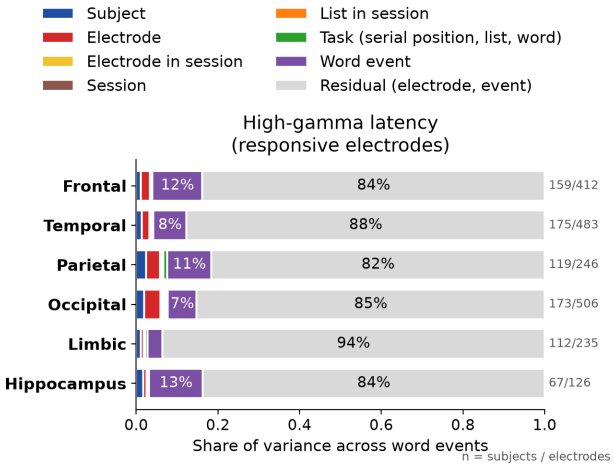



**Figure 3:** (A) High-gamma half-max latency per region. (B) Variance components of latency per word event (responsive electrodes).

(Done)

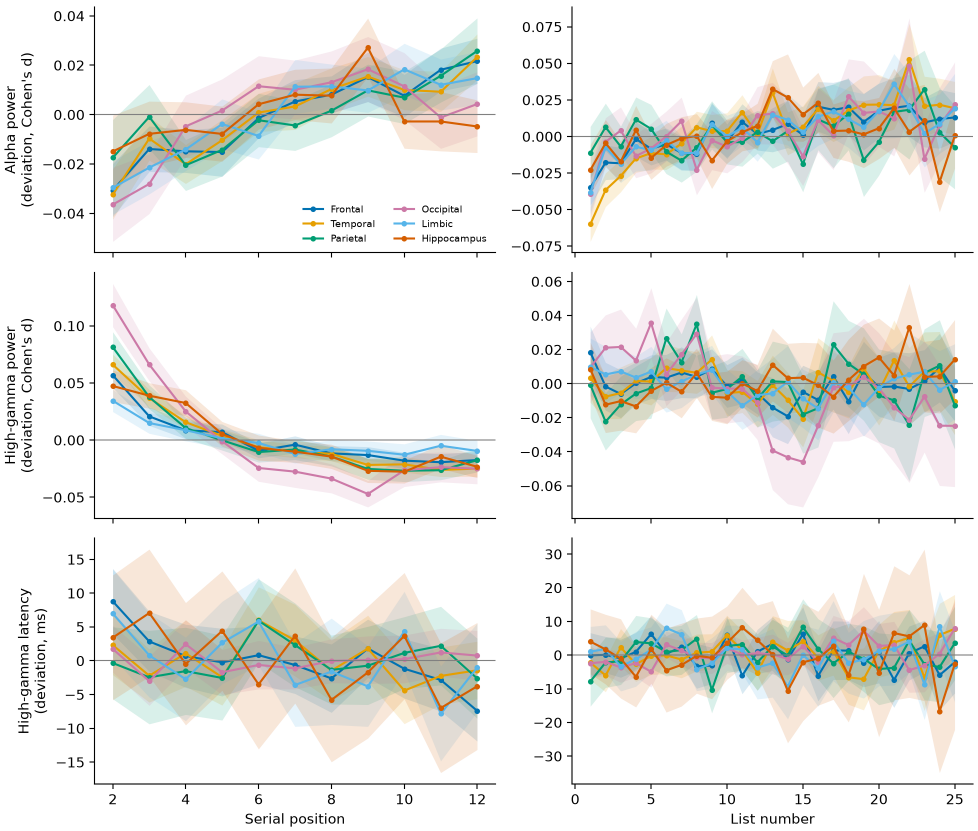
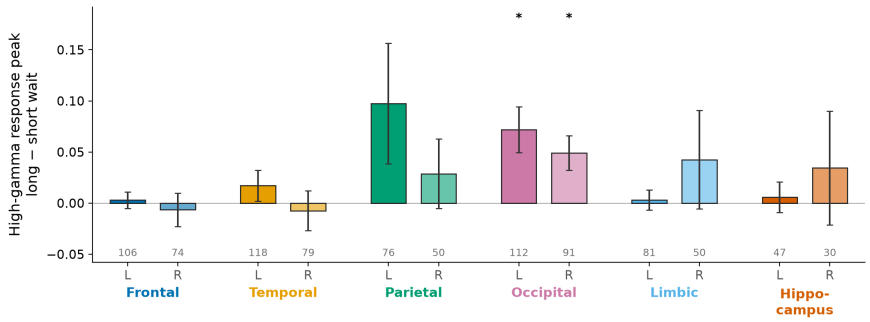



**Figure 4:** Serial-position and list effects: deviation from each subject's own mean (all electrodes; latency: responsive electrodes); mean and 95% CI across subjects.

**A** (Done)




**B** (Planned)



**C** (Planned)

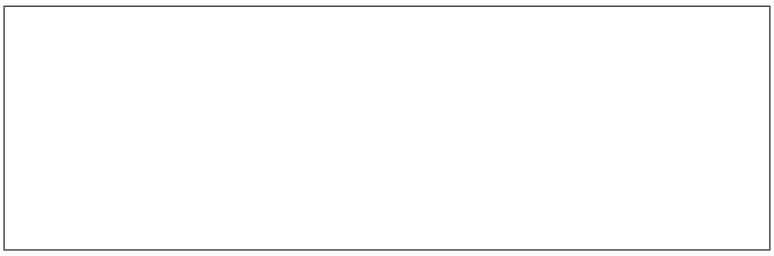
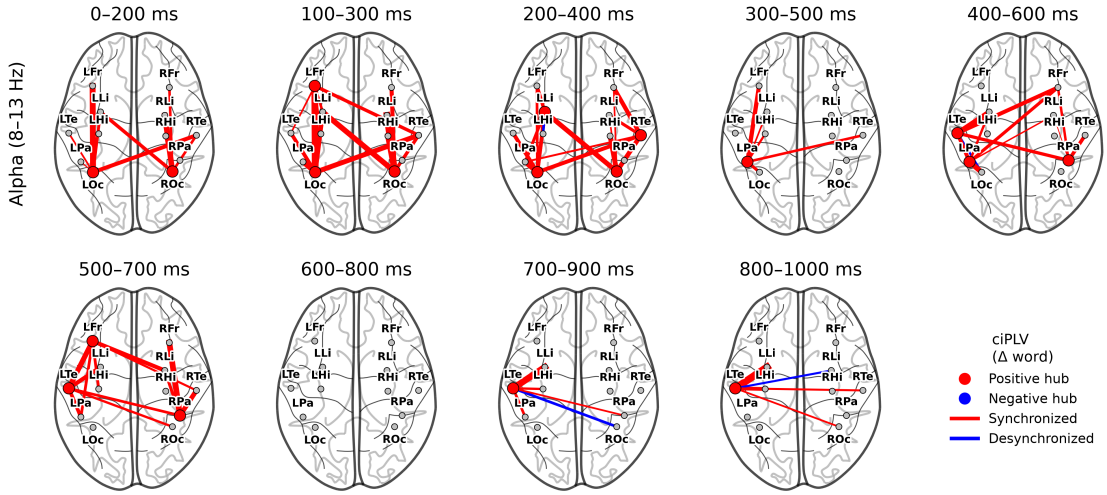
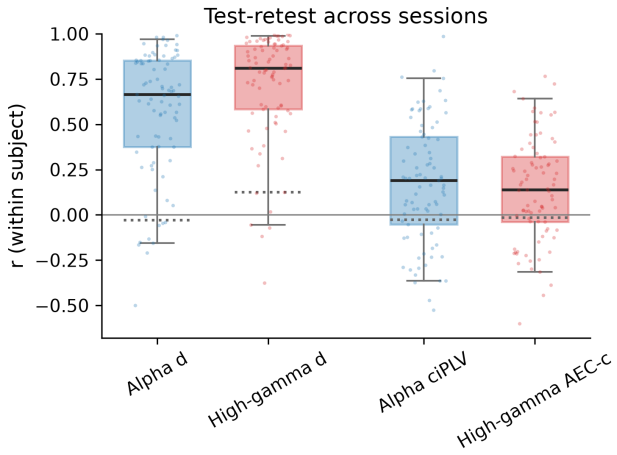


**Figure 5:** (A) High-gamma response peak after long vs. short blanks, per region (stars: FDR; n: subjects). (B) Effect of the previous word's response on the current word, by blank length. (C) Variance components of power in the pre-stimulus blank (Aim 1's split).

**A** (Done)




**B** (Done)




**C** (Planned)



**Figure 6:** (A) Alpha ciPLV change at word onset in 200 ms windows: hubs (nodes) and each hub's five strongest connections. (B) Test-retest across sessions: $r$ across regions between halves of a subject's sessions (dots: subjects). (C) Variance components of synchrony.

## References


Burke, J. F., Long, N. M., Zaghloul, K. A., Sharan, A. D., Sperling, M. R., and Kahana, M. J. (2014). Human intracranial high-frequency activity maps episodic memory formation in space and time. *NeuroImage*, 85:834–843.

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